

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer, adept at building scalable microservices and distributed data pipelines. Possesses a Master of Science in Computer Science with expertise in Java, Python, Spark, SQL, AWS, and Azure, complemented by experience in CI/CD automation, large-scale data platforms, and privacy-preserving AI applications. Seeking to leverage expertise in distributed systems and software engineering to excel as a Backend Software Engineer in Trading & Data Platforms.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Architected and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, managing over 50,000 daily API requests with 99.99 percent SLA uptime.
- Developed and automated CI/CD pipelines utilizing Jenkins, Maven, and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments, enhancing deployment speed by 30%.
- Spearheaded the modernization of legacy systems by migrating to Log4j 2.x and enhancing monitoring and error handling, decreasing issue resolution time by 20 percent.
- Led a significant EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions utilizing Git, SVN, and JUnit testing, ensuring robust software engineering practices.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, improving system throughput.
- Enhanced system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code for critical financial applications.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, reducing integration defects by 15%.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering team growth in software engineering principles.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization, ensuring critical data platform stability.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity for fintech solutions.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and enhancing software quality.
- Supported release planning and sprint execution using JIRA within Agile development teams, contributing to efficient workflow management.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Docker, Terraform, AWS) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real-time, enabling critical data ingestion for analytics.
- Engineered an interactive Flask web application for data exploration with dynamic filtering and visualization, providing actionable insights.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments, ensuring efficient workflows.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent and enhancing distributed systems performance.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, supporting scalable data platforms.

Privacy-Preserving AI Email Client (Capstone Project - T5, LoRA, Hugging Face) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, demonstrating AI tool application.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, relevant for sensitive data handling.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, showcasing expertise in privacy-preserving AI and distributed systems concepts.

SKILLS

- **Programming Languages:** Proficient in Java, Python, and SQL for robust software engineering and data analysis.
- **Frameworks & Libraries:** Extensive experience with Spring Boot, Apache Spark, PySpark, Hugging Face, T5, and LoRA for building scalable applications and AI tools.
- **Web Services:** Skilled in designing and implementing REST API and SOAP API, with exposure to gRPC and GraphQL for efficient distributed systems communication.
- **Cloud Platforms:** Expertise in AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and GCP, supporting infrastructure automation and cloud-native development.
- **Databases:** Experienced with PostgreSQL, Oracle Database, IBM DB2, and RDS for managing diverse data platforms.
- **Tools & Technologies:** Proficient in Docker/Kubernetes, Jenkins, Git, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for CI/CD, workflow management, and infrastructure automation.
- **Concepts:** Strong understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, Incident Response, and Workflows.
- **Data & AI:** Expertise in Data Ingestion, Machine Learning, AI Tools, Fintech applications, and advanced Distributed Systems for trading platforms.